

# The Art Gallery Problem

14 de noviembre de 2025

## Definition (The art gallery problem AGP)

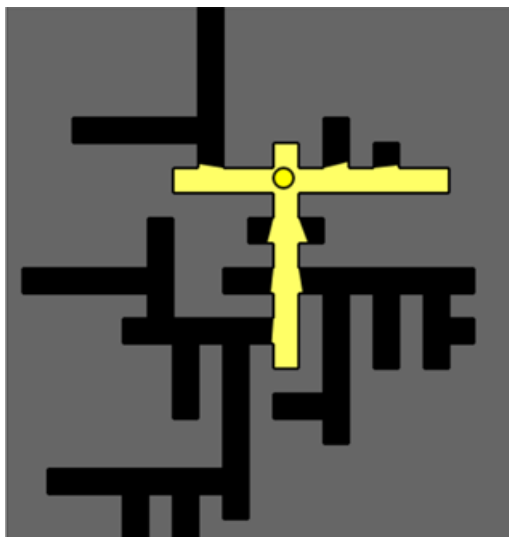
The art gallery problem or museum problem is a well-studied visibility problem in computational geometry.

It originates from the following real-world problem:

*"In an art gallery, what is the minimum number of guards who together can observe the whole gallery?"*

In the geometric version of the problem, the layout of the art gallery is represented by a simple polygon and each guard is represented by a point in the polygon. A set

$S$  of points is said to guard a polygon if, for every point  $p$  in the polygon, there is some  $q \in S$  such that the line segment between  $p$  and  $q$  does not leave the polygon.



## Definition (3-SAT)

Determining the satisfiability of a formula in conjunctive normal form where each clause is limited to at most three literals is called 3-SAT, 3CNFSAT, or

3-satisfiability.

### Theorem (Cook-Levin)

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| 3-SAT is NP-hard |
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*Demonstration:* Trivial, wikipedia

### Definition (3-coloring problem)

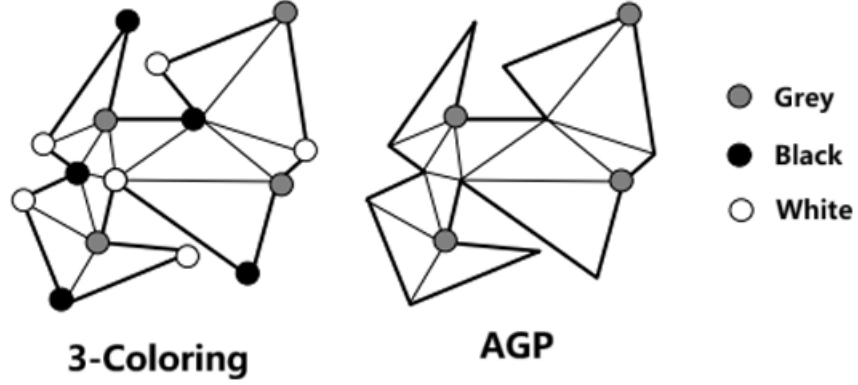
The 3-coloring problem is a decision problem in graph theory that asks whether the vertices of a given graph can be colored using at most three colors such that no two adjacent vertices share the same color

### Lemma

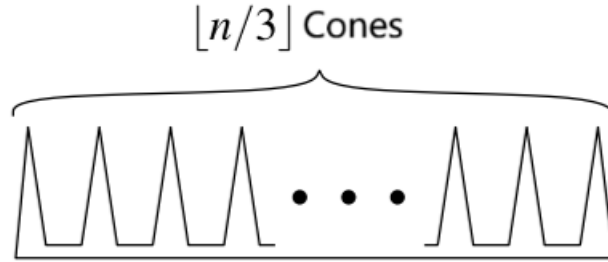
|                                                                                                 |
|-------------------------------------------------------------------------------------------------|
| An AGP can be polynomial-time reduced to 3-Coloring, or $3 - \text{Coloring} \leq_p \text{AGP}$ |
|-------------------------------------------------------------------------------------------------|

*Demonstration:* Let TP be a triangulation of P. Select a subset of the vertices of P such that every triangle in TP has at least one selected vertex, and place cameras at the selected vertices. To find such a subset, we assign each vertex of P a color: white, gray, or black. The coloring will be such that any two vertices connected by an edge or a diagonal have different colors. This is called a 3-coloring of a triangulated polygon. In a 3-coloring of a triangulated polygon, each triangle has one white, one gray, and one black vertex.

Therefore, if we place cameras on all gray vertices, for example, we have protected the entire polygon. Since each triangle must have all three colors on its three vertices, every triangle has a gray node on one of its vertices. Suppose we place cameras on every gray node. Then every triangle has a camera on one of its vertices. Clearly, a triangle is covered by a camera located at one of its vertices. Thus, every triangle is covered. Finally, the collection of triangles in a triangulation completely covers the polygon. Therefore, the entire polygon is covered if cameras are placed on the gray nodes. Similarly, the entire polygon is covered if cameras are placed on the white, gray, or black nodes. Which gives a optimized solution of  $\lfloor \frac{n}{3} \rfloor$



On the other hand,  $\lfloor \frac{n}{3} \rfloor$  is the best solution since the counterexample of:



Thus we conclude that we have established a polynomial-time reduction from AGP to 3-Coloring.

### Lemma

3-Coloring problem can be polynomial-time reduced to 3-SAT, or  $3-Coloring \leq_p 3-SAT$

*Demonstration:*

First let  $v_R$  be *True* if vertex  $v$  is colored Red, and same for  $v_G$  and  $v_B$ .

And following the rules of 3-coloring, we shall define:

1. At least one color:  $v_R \vee v_G \vee v_B$
2. At most one color:  $(\neg v_R \vee \neg v_G) \wedge (\neg v_G \vee \neg v_B) \wedge (\neg v_B \vee \neg v_R)$
3. No same color on one edge:  $\forall (u, v) \in Edge \quad (\neg v_R \vee \neg u_R) \wedge (\neg v_G \vee \neg u_G) \wedge (\neg v_B \vee \neg u_B)$

Therefor we have our 3-SAT defined, thus we conclude that we have established a polynomial-time reduction from 3-Coloring to 3-SAT.

### Lemma

AGP is NP-hard

*Demonstration:* Chain thorem and by definition of NP-hard, trivial